ASHIVAM TECHNOLOGIES

INFINITE POSSIBILITIES. ENGINEERED WITH PRECISION.

CHAPTER 1 • DOCUMENT CONTROL & HANDOVER BLUEPRINT

Official Website Project Documentation

A comprehensive architectural, technical, operational, and security guide for the Ashivam Technologies full-stack digital platform.

DOCUMENT VERSION
Version 2.2 (Audited Handover Blueprint)

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02 Executive Summary

2.1 Project Overview & Business Purpose

The **Ashivam Technologies Official Website** serves as the central digital flagship and enterprise engagement platform for Ashivam Technologies. Designed as an enterprise-grade web application, it bridges client acquisition, brand authority, talent recruitment, and internal employee document operations into a cohesive, high-performance architecture.

The system is built upon modern full-stack web standards: a **React 18 single-page application (SPA)** powered by **Vite 8** on the client tier, coupled with a hardened, modular **Express 5.2 REST API** backed by **MongoDB Atlas** on the server tier. The user interface embodies a high-contrast dark enterprise aesthetic accented with refined gold gradients, engineered to convey technical rigor, stability, and institutional reliability.

2.2 Key Stakeholders & Target User Personas

- **Enterprise B2B Clients & Partners:** Exploring capabilities across custom software, cloud architecture, native mobile applications, and UI/UX design. They interact via the dynamic inquiry portal and sector-specific case study showcases.
- **Engineering Candidates & Job Applicants:** Browsing published technical openings across Android, backend, QA, and frontend roles, submitting applications with PDF/Word resume uploads directly through the public careers gateway.
- **System Administrators & Company Leadership:** Authenticating via dedicated administrative gateways to manage job listings, review inbound project inquiries, inspect recruitment pipelines, and manage corporate settings.
- **Internal Employees & HR Personnel:** Utilizing the employee document and HR management modules for internal operational records and credential verification.

2.3 Handover Status & Milestone Verification Scope

As of October 2026, the project has completed extensive UI/UX optimization, dead-space elimination, and Git source-code transfer to company ownership. **Important Scope Distinction:** Source code custody has been transferred and synchronized to the company GitHub repository; provisioning of company-owned Render, Vercel, MongoDB Atlas, and AWS/Cloudflare accounts remains pending.

MILESTONE DIMENSION	VERIFIED STATUS	EVIDENCE / LOCATION / CONSTRAINTS
Source Code Custody	SYNCHRONIZED	Transferred to official company repository: `https://github.com/AshivamTechnologies/Official-Websites.git` at commit `6b2fae02b0aca4138fb4ecb77d7b7cfe744ae408`.
Frontend Test Suite	65 PASSED / 0 FAILED	Static & unit regression suite: routes, 44px touch targets, mobile drawer, responsive clamp typography, and SEO (828ms).
Backend Test Suite	79 PASSED / 0 FAILED	Unit & supertest HTTP integration suite: validation, auth guards, rate limiters, error envelopes, and proxy trust (5,858ms).
Production Build Integrity	VERIFIED	Vite production build bundles 86 modules into `dist/` in 280ms without errors or warnings.
Current Live Frontend	PRESERVED (PERSONAL)	Hosted on personal Vercel deployment (`https://ashivam-tec-h.vercel.app/`) serving release bundle `index-0vra-z_U.js`. Preserved during transition; current uptime unverified.
Current Live Backend	PRESERVED (PERSONAL) / UNVERIFIED HEALTH	Deployment URL is preserved on personal Render account (`https://ashivam-tech.onrender.com/api/v1`), but current service/database health was not verified during this audit (health probe timed out on free-tier spin-down).
Company Cloud Infrastructure (Target)	PENDING PROVISIONING	Target architecture: company-owned Render, Vercel, MongoDB Atlas, AWS S3/R2, and Brevo accounts have not been provisioned yet. Handover documents target specifications only.

03 Project Scope & Implemented Feature Catalog

3.1 Public Client-Facing Pages & Workflows

1. Homepage (`/`)

- **Hero Section:** High-impact value proposition ("Infinite Possibilities. Engineered with Precision."), dual call-to-action buttons ("Discuss Your Project", "Explore Solutions"), and animated background glow elements.
- **Institutional Metrics (`Stats`):** Displays operational metrics from marketing copy (active projects, developer contributors, supported technologies) — *[Marketing copy claim — Requires company verification / corporate metrics sign-off]*.
- **Core Capabilities (`Services`):** Interactive 4-card matrix detailing Web Development, Mobile Development, Custom Software Engineering, and UI/UX Design Systems.
- **Agile Delivery Lifecycle (`Process`):** A modern 4-phase interactive process section ("Discovery & Architecture", "Rapid Prototyping", "Production Hardening", "Deployment & Handover") with numbered indicators, status tags, and deliverable badges.
- **Client Inquiry & Lead Capture (`Contact`):** High-conversion contact form featuring real-time client-side regex validation, phone capture, silent honeypot anti-spam defense, project budget tiers, and instant confirmation feedback.
- **Global Sticky Custom Cursor:** Custom gold interactive cursor implemented via `useCursor.js`, featuring hover physics on interactive targets and leak-proof lifecycle listeners.

2. Dedicated Industries Portal (`/industries` , `/industries/:sector`)

- Streamlined single-header layout eliminating legacy double-stacked headings.
- Interactive sector filter pills (Fintech, Healthtech, Enterprise SaaS, E-Commerce, Logistics, Edtech) with minimum 44px touch targets.
- Six domain architecture cards detailing regulatory compliance, architecture deliverables, and technical stack tags.

3. Dedicated Solutions Portal (`/solutions` , `/solutions/:category`)

- Modular category filter bar navigating Architecture, Frontend, Cloud Backend, UI/UX, and Storage.
- Compact SaaS-inspired service cards featuring capability checklists and technology pill badges.
- Embedded engineering case studies showcase (`Projects`) detailing delivered client solutions.

4. Careers & Recruitment Gateway (`/careers`)

- Dynamic recruitment board fetching published job openings from `/api/v1/career/jobs`.
- Automatic filtering excluding draft or expired listings from public view.
- Interactive application modal supporting applicant detail capture and resume upload (.pdf, .doc, .docx up to 5MB).

5. Legal Governance & Compliance Suite (`/privacy-policy` , `/terms-of-service` , `/security`)

- Unified responsive legal container (`LegalPage.jsx`) with persistent tab navigation.
- **Privacy Policy:** Zero-sale personal data commitment, inquiry data custody, and statutory privacy rights.
- **Terms of Service:** Engineering governance, 100% client IP assignment upon milestone payment, mutual NDA clauses, and Agra, UP judicial jurisdiction — *[Public website copy claim — Requires company verification / corporate legal review]*.
- **Security Statement:** Input parameterization, cryptographic JWT sessions, rate limiting, and 24-business-hour vulnerability disclosure SLA — *[Public website copy claim — Requires company verification / corporate legal review]*.

3.2 Administrative & Internal Capabilities

- **Dual Portal Selector (`/employee-login`):** Gateway routing staff to Employee Login or Administrator Login.
- **Admin Authentication (`/admin/login` , `/employee-login/admin`):** Cryptographically secured credential login issuing signed JWT tokens with 1-hour expiration and session-storage persistence.
- **Full-Stack Admin Dashboard (`/admin`):** Modular administrative panel managing Inquiries, Job Postings, Candidate Applications, Employees, HR Documents, and Site Settings.
- **Dynamic Job Management:** Real-time job creation, slug generation, activation/deactivation toggles, and applicant resume streaming.

04 Comprehensive Technology Stack Matrix

Every library and runtime dependency across the root frontend and backend applications has been verified directly against `package.json` and lockfiles.

4.1 Frontend Tier Technologies (Root Workspace)

TECHNOLOGY / PACKAGE	VERIFIED VERSION	CLASSIFICATION	ARCHITECTURAL RESPONSIBILITY
React	`^18.3.1`	Runtime Dependency	Core declarative component library and Virtual DOM engine.
React DOM	`^18.3.1`	Runtime Dependency	DOM renderer for React web application.
React Router DOM	`^7.18.4`	Runtime Dependency	Declarative client-side routing, route guards, and history management.
Vite	`^8.3.4`	Build / Dev Tool	Next-generation frontend dev server and production Rollup bundler.
@vitejs/plugin-react	`^6.1.2`	Build Tool	Babel/Fast-refresh integration for React JSX transformation.
esbuild	`^0.25.0`	Package Override	Extremely fast JS/TS compilation and minification engine.
CSS3 Custom Properties	Native	Styling Architecture	Design token system, responsive clamp typography, and theme variables.

4.2 Backend Tier Technologies (`backend/`)

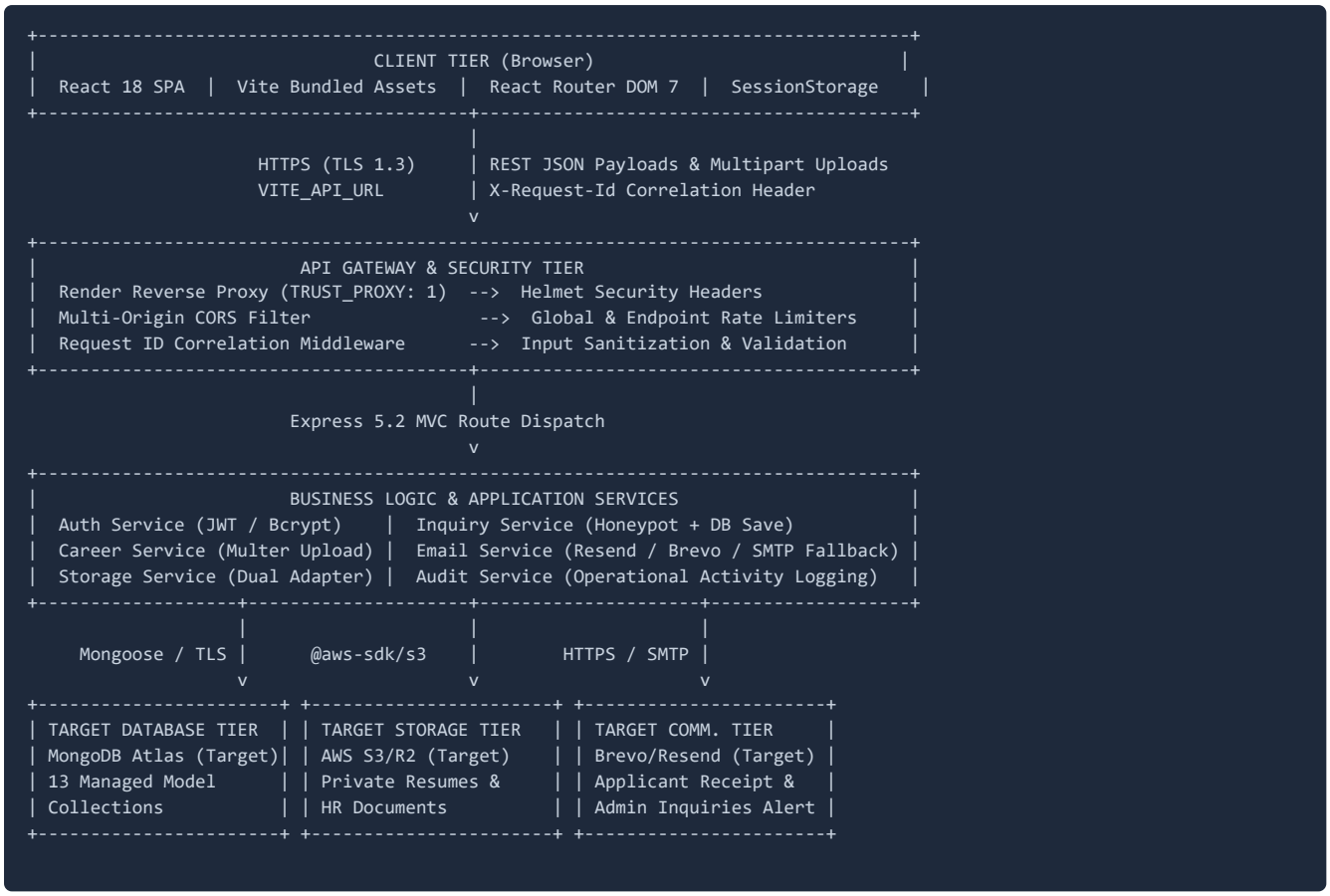
TECHNOLOGY / PACKAGE	VERIFIED VERSION	CLASSIFICATION	ARCHITECTURAL RESPONSIBILITY
Node.js	`v22.23.1` (LTS)	Server Runtime	High-concurrency asynchronous JavaScript runtime engine.
Express	`^5.2.1`	Runtime Dependency	Modern HTTP server framework routing APIs, middleware, and errors.
Mongoose	`^9.10.3`	Runtime Dependency	Object Data Modeling (ODM) for MongoDB with schema validation.
Helmet	`^8.3.0`	Security Middleware	Enforces essential HTTP security headers (HSTS, Sniff, Framing).
CORS	`^2.8.6`	Security Middleware	Restricts cross-origin resource sharing to authorized frontend domains.
express-rate-limit	`^8.7.0`	Security Middleware	Multi-tiered IP rate limiter defending against DDoS and brute-force.
jsonwebtoken	`^9.0.3`	Auth Dependency	Generates and verifies cryptographic HMAC SHA-256 session tokens.
bcryptjs	`^3.0.3`	Security Dependency	One-way salt-hashed password protection for administrator records.
multer	`^2.4.0`	File Middleware	Multipart form-data parser for inbound applicant resume uploads.
@aws-sdk/client-s3	`^3.1148.0`	Cloud Storage SDK	Official AWS SDK client for S3/R2. Implemented in code; target company bucket pending setup *[Requires company verification]*.
nodemailer	`^10.0.13`	Email Transport	SMTP and HTTPS transport engine (Resend/Brevo/SMTP). Implemented in code; company credentials pending setup *[Requires company verification]*.
dotenv	`^18.0.5`	Environment Tool	Loads local configuration variables from `.env` files in development.
supertest	`^7.3.1`	Dev / Test Dependency	HTTP assertions for end-to-end integration test execution.
nodemon	`^3.1.14`	Dev Dependency	Local development server file watcher with automatic restarts.
chokidar	`^4.0.3`	Package Override	High-performance cross-platform file watcher override.

05 Architecture, Component Topography & Data Flows

5.1 Full-Stack System Architecture

The platform follows an asynchronous, decoupled client-server architecture. In target production architecture, the frontend is designed for Vercel's global edge network as pre-compiled static assets, while the backend operates as a stateless Express container on Render communicating over TLS with target cloud services (MongoDB Atlas, AWS S3/R2, and Brevo/Resend). Note: Current personal developer deployments are preserved, while target company cloud provisioning remains pending.

Figure 5.1 — High-Level End-to-End System Topography



5.2 Core Transactional Sequence Flows

1. Client Inquiry Submission Flow

1. **Client Input:** User inputs name, email, phone, project category, and message on `/` or `/contact`.
2. **Client-Side Validation:** Form verifies email format, phone digits (7-20 characters), and message length (≥ 10 characters).
3. **Payload Dispatch:** `fetch()` dispatches JSON payload to `POST /api/v1/inquiries` with `X-Request-Id`.
4. **Rate Limiting Guard:** `publicFormLimiter` validates that client IP has not exceeded 10 requests per 15 minutes.
5. **Honeypot Inspection:** Backend validator checks hidden bot trap field; bots are silently acknowledged with 200 without saving.
6. **Persistence:** Sanitized inquiry is written to MongoDB Atlas `inquiries` collection.
7. **Notification Dispatch:** Asynchronous email notification triggered via configured provider (Resend, Brevo, or SMTP) to `NOTIFICATION_RECIPIENT`, plus receipt acknowledgment to applicant.
8. **Response:** Returns standardized JSON envelope `{ success: true, message: "Inquiry received" }` (HTTP 201).

2. Career Application & Resume Storage Flow

1. **Candidate Submission:** Applicant fills modal form at `/careers` and selects PDF/DOC resume file.
2. **Multer Interception:** Backend `multer` memory-buffer middleware captures multipart payload (max 5MB).
3. **Magic-Byte Verification:** `validateFileSignature()` in `storage.service.js` inspects the first 4 bytes of the binary buffer to verify authentic PDF (`%PDF`) or DOC/DOCX signatures, defeating malicious file extension spoofing.
4. **Storage Adapter Routing:**
   - If `STORAGE_PROVIDER === 's3'`: Streams buffer to private S3 bucket using `PutObjectCommand` with generated UUID key and AES256 encryption.
   - If `STORAGE_PROVIDER === 'local'`: Writes file to isolated local disk folder `uploads/resumes/`.
5. **Database Record:** New `Application` document is stored in MongoDB with applicant contact details and `resume: { storageProvider, storageKey, fileName, size }`.

6. **Access Protection:** Uploaded resumes are explicitly blocked from public HTTP access via Express route exclusion. Admin access requires authenticated Bearer JWT token to generate authorized streaming response.

06 Complete Project Directory Anatomy

The project follows a clean monorepo-style structure where frontend client assets reside at the root and server architecture is encapsulated within the `backend/` directory.

```
C:\Users\DELL\Desktop\Official-Website-main\
├── .gitignore                    # Git exclusion rules (node_modules, .env*, dist, uploads)
├── .env.example                  # Frontend environment configuration template
├── index.html                    # Single-page application root HTML entry with JSON-LD
├── package.json                  # Frontend dependencies, build, and test scripts
├── render.yaml                   # Infrastructure-as-code deployment manifest for Render
├── vercel.json                   # Vercel SPA routing rewrites and security headers
├── vite.config.js                # Vite bundler, plugin, and server configuration
│
├── src/                          # Frontend React Source Directory
│   ├── main.jsx                  # Application DOM mount entry point
│   ├── App.jsx                   # Global router defining all 17 public & admin routes
│   ├── api.js                    # Centralized API client with JWT token session handling
│   ├── components/               # Reusable modular UI components (Navbar, Hero, Stats, Process, etc.)
│   ├── pages/                    # Dedicated multi-page route views (Industries, Solutions, Careers, Legal)
│   ├── data/content.js           # Centralized company data, social links, and navigation
│   ├── hooks/useCursor.js        # Custom interactive cursor hook with cleanup lifecycle
│   ├── styles/                   # Global stylesheets, animation keyframes, and tokens
│   └── utils/                    # Dynamic SEO metadata updater and GA4 analytics
│
├── backend/                      # Express REST API Server Workspace
│   ├── package.json              # Backend dependencies, test scripts, and seed/migration runners
│   ├── Dockerfile                # Production container specification for portable cloud hosts
│   ├── .env.example              # Backend environment variable template
│   ├── src/
│   │   ├── server.js             # Process bootstrap, unhandled rejection hooks, port listener
│   │   ├── app.js                # Express app, Helmet, CORS, rate limiters, route mounting
│   │   ├── config/env.js         # Strict environment validator, origin parser, defaults
│   │   ├── db/database.js        # Mongoose connection pooling and error handling
│   │   ├── models/admin.model.js # Admin credential and RBAC Mongoose model
│   │   ├── middlewares/          # Request correlation, logging, auth guards, rate limiters
│   │   ├── modules/              # Domain modules (auth, career, inquiry, employee, settings)
│   │   ├── scripts/              # create-admin.js, migrate-resumes-to-s3.js, verify-smtp.js
│   │   ├── services/storage.service.js # Dual S3 / local disk storage adapter with magic-byte audit
│   │   └── utils/                # JWT helpers, input sanitization, structured JSON logger
│   └── tests/                    # 79 automated backend unit & HTTP integration tests
│
└── tests/                        # 65 automated frontend component & SEO regression tests
```

SCRIPT EXECUTION SCOPE NOTICE

Backend administrative scripts — including `npm run migrate:resumes`, `npm run seed`, and `npm run create:admin` — are declared in `backend/package.json`. They must be executed from within the `backend/` directory (`cd backend && npm run ...`) or targeted via `npm --prefix backend run ...`.

07 Prerequisites & Environmental System Requirements

7.1 Supported Operating Systems & Runtimes

SYSTEM COMPONENT	MINIMUM REQUIRED	RECOMMENDED PRODUCTION	DIAGNOSTIC VERIFICATION COMMAND
Node.js Runtime	`v20.0.0`	`v22.23.1` (LTS)	`node --version`
Node Package Manager	`npm 10.0.0`	`npm 10.9.8`	`npm --version`
Git Version Control	`git 2.30.0`	`git 2.55.0+`	`git --version`
MongoDB Database	`v6.0.0`	MongoDB Atlas Cluster (v7+)	`mongosh --eval "db.version()"`
Operating System	Windows 10/11, macOS 12+, Ubuntu 22.04 LTS, or Debian 12	Linux (Debian/Alpine Docker container)	`uname -a` (Linux/macOS) or `winver` (Windows)

08 Step-by-Step Installation & Local Setup Guide

Follow these exact copy-paste sequences in your terminal to initialize and run the complete full-stack platform locally.

8.1 Dual Platform Commands (PowerShell & Bash)

1. Clone Repository & Navigate to Workspace

```
# PowerShell / Windows:
git clone https://github.com/AshivamTechnologies/Official-Websites.git
cd Official-Websites

# Bash / Linux / macOS:
git clone https://github.com/AshivamTechnologies/Official-Websites.git
cd Official-Websites
```

2. Install Root Frontend & Backend Dependencies

> **IMPORTANT DEPENDENCY ISOLATION**
> Root dependencies and backend dependencies are decoupled and must be installed separately. Do not run `npm install` only in root.

```
# Install Frontend Dependencies (Root Workspace):
npm install

# Install Backend Dependencies:
cd backend
npm install
cd ..
```

3. Initialize Environment Files

```
# Create Frontend Local Environment:
Copy-Item .env.example .env.local    # PowerShell
cp .env.example .env.local           # Bash

# Create Backend Local Environment:
Copy-Item backend/.env.example backend/.env    # PowerShell
cp backend/.env.example backend/.env           # Bash
```

4. Start Backend Server (Terminal 1)

```
cd backend
npm run dev
# Expected Output: [Ashivam Backend] Operational on port 5000 (development)
# [Database] MongoDB connected successfully
```

5. Start Frontend Dev Server (Terminal 2)

```
npm run dev
# Expected Output: VITE v8.3.4 ready in 240 ms
# ➜  Local:   http://localhost:5173/
```

09 Complete Environment Variable Reference

ZERO-SECRET POLICY & COMPLIANCE

This chapter documents environment variable names, defaults, and requirements strictly using placeholders. In compliance with security standards, no passwords, API keys, MongoDB Atlas connection strings, or JWT secrets are exposed.

9.1 Frontend Environment Variables (`.env.local` / Vercel)

VARIABLE NAME	STATUS	DEFAULT IN CODE	FEATURE AFFECTED	RESTART REQUIRED?
`VITE_API_URL`	MANDATORY	`https://ashivam-tech.onrender.com/api/v1`	Configures API destination in `src/api.js` for all public forms and admin dashboards.	Yes (Frontend rebuild required)
`VITE_GA_ID`	OPTIONAL	`undefined`	Google Analytics 4 Measurement ID in `src/utils/analytics.js`.	Yes (Frontend rebuild required)

9.2 Backend Environment Variables (`backend/.env` / Render)

VARIABLE NAME	STATUS	DEFAULT IN CODE	PURPOSE & SECURITY CONSTRAINT	RESTART REQUIRED?
`NODE_ENV`	MANDATORY	`development`	Controls production security enforcement in `env.js`. Must be `production` in live hosts.	Yes (Process restart)
`PORT`	OPTIONAL	`5000` (Local) / `10000` (Render)	TCP port bound by HTTP server listener.	Yes (Process restart)
`CLIENT_URL`	MANDATORY	`http://localhost:5173` (Dev) / Vercel Fallback	Canonical frontend URL used to construct links in transactional emails.	Yes (Process restart)
`ALLOWED_ORIGINS`	MANDATORY	Localhost + Vercel domains	Comma-separated whitelist of allowed CORS origins.	Yes (Process restart)
`MONGODB_URI`	MANDATORY	`mongodb://127.0.0.1:27017/...`	MongoDB connection string. **Must be Atlas URI in production** (rejects localhost).	Yes (Process restart)
`JWT_SECRET`	MANDATORY	Dev fallback (Rejected in Prod)	HMAC secret for admin JWT tokens. **Must be ≥ 32 characters in production**.	Yes (Process restart)
`JWT_EXPIRES_IN`	OPTIONAL	`1h`	Validity duration for issued admin access tokens.	Yes (Process restart)
`TRUST_PROXY`	OPTIONAL	`1` (Prod) / `false` (Dev)	Express reverse proxy hop count for accurate client IP resolution behind Render.	Yes (Process restart)
`EMAIL_PROVIDER`	OPTIONAL	`resend`, `brevo`, or `smtp`	Selects primary email delivery driver (auto-detects from API keys).	Yes (Process restart)
`RESEND_API_KEY`	CONDITIONAL	`""`	API key for Resend HTTPS delivery (bypasses cloud SMTP port blocks).	Yes (Process restart)
`BREVO_API_KEY`	CONDITIONAL	`""`	API key for Brevo HTTPS delivery alternative.	Yes (Process restart)
`SMTP_HOST`	CONDITIONAL	`""`	SMTP gateway host (e.g. `smtp.sendgrid.net` or Google Workspace).	Yes (Process restart)
`SMTP_PORT`	CONDITIONAL	`587`	SMTP port (587 for STARTTLS, 465 for SSL).	Yes (Process restart)
`SMTP_USER`	CONDITIONAL	`""`	Authentication username for SMTP gateway.	Yes (Process restart)
`SMTP_PASSWORD`	CONDITIONAL	`""`	Authentication secret / application token for SMTP gateway.	Yes (Process restart)
`EMAIL_FROM`	MANDATORY	`Ashivam Technologies <no-reply...>`	Verified sender email header displayed to recipients.	Yes (Process restart)
`NOTIFICATION_RECIPIENT`	MANDATORY	`admin@ashivamtechnologies.com`	Company inbox where inbound lead inquiries and job applications are alerted.	Yes (Process restart)

VARIABLE NAME	STATUS	DEFAULT IN CODE	PURPOSE & SECURITY CONSTRAINT	RESTART REQUIRED?
`STORAGE_PROVIDER`	MANDATORY (PROD)	`local` (Dev) / `s3` (Prod)	Object storage mode (`s3` required for non-ephemeral storage in cloud).	Yes (Process restart)
`S3_BUCKET`	CONDITIONAL	`""`	Target AWS S3 or Cloudflare R2 bucket name. Mandatory if storage is `s3`.	Yes (Process restart)
`S3_REGION`	CONDITIONAL	`us-east-1`	S3 bucket region (e.g. `ap-south-1` for Mumbai, or `auto` for R2).	Yes (Process restart)
`S3_ACCESS_KEY_ID`	CONDITIONAL	`""`	IAM programmatic user access key. Mandatory if storage is `s3`.	Yes (Process restart)
`S3_SECRET_ACCESS_KEY`	CONDITIONAL	`""`	IAM programmatic user secret access key. Mandatory if storage is `s3`.	Yes (Process restart)

10 Database Architecture, Schema Models & Safe Migration Runbook

10.1 Database Connection Configuration

The target database tier is architected for MongoDB Atlas and managed via Mongoose in `backend/src/db/database.js`. Note: Company MongoDB Atlas cluster provisioning and live database connectivity remain pending verification. The connection engine employs production-grade pooling parameters:

- `maxPoolSize: 10` and `minPoolSize: 2` maintain warm socket pools while preventing cloud connection limits from exhausting.
- `serverSelectionTimeoutMS: 5000` guarantees fast failure recovery rather than hanging requests during cloud network blips.
- `autoIndex: !env.isProduction` disables automatic runtime index builds in production, preventing unexpected performance degradation on Atlas clusters during boot.

10.2 Complete Collection Schema Inventory (13 Models)

MODEL NAME	COLLECTION NAME	SOURCE MODEL FILE	CORE SCHEMA FIELDS & VALIDATION
Admin	`admins`	`models/admin.model.js`	`username` (unique index), `name`, `email`, `passwordHash`, `role` (Superadmin, Admin), `permissions`, `isActive`, `lastLoginAt`.
Application	`applications`	`career/application.model.js`	`jobId`, `fullName`, `email`, `phone`, `experience`, `resume: { storageProvider, storageKey, fileName, size }`, `status` (pending, reviewed, rejected, hired).
Job	`jobs`	`career/job.model.js`	`title`, `slug` (unique index), `department`, `location`, `type`, `experience`, `description`, `requirements`, `status` (published, draft, closed).
Inquiry	`inquiries`	`inquiry/inquiry.model.js`	`name`, `email`, `phone`, `company`, `projectType`, `budget`, `message`, `status` (new, in-review, contacted, archived).
Employee	`employees`	`employee/employee.model.js`	`employeeId`, `firstName`, `lastName`, `email`, `phone`, `role`, `department`, `designation`, `status` (active, inactive, on-leave).
EmployeeDocument	`employee_documents`	`employee-document/...`	`employeeId`, `title`, `documentType`, `fileUrl`, `storageKey`, `verified`.
HrDocument	`hr_documents`	`hr-document/...`	`title`, `category`, `version`, `description`, `fileUrl`, `storageKey`, `restrictedRoles`.
Internship	`internships`	`internship/internship.model.js`	`title`, `domain`, `duration`, `stipend`, `requirements`, `active`.
AuditLog	`audit_logs`	`audit/audit.model.js`	`userId`, `action`, `resource`, `resourceId`, `ipAddress`, `userAgent`, `timestamp`.
Settings	`settings`	`settings/settings.model.js`	`company: { name, email, phone, address }`, `social: { linkedin, github }`, `maintenance: { enabled, message }`.
Media	`media`	`media/media.model.js`	`fileName`, `mimeType`, `size`, `url`, `uploadedBy`, `publicAccess`.
Notification	`notifications`	`notification/...`	`recipient`, `subject`, `body`, `type`, `read`, `sentAt`.
Content	`contents`	`content/content.model.js`	`key` (unique), `title`, `body`, `metadata`, `updatedBy`.

10.3 Safe Isolated-Target Migration Protocol (Non-Destructive)

> **CRITICAL DATA SAFETY NOTICE: --DROP IS DESTRUCTIVE**
>
> **Never execute `mongorestore` with the `--drop` flag against production or active databases.** The `--drop` flag purges every existing collection before restoring. The standard migration workflow must always target a fresh, isolated company database cluster without dropping collections.

Standard Isolated-Target Migration Workflow

```
# 1. Non-destructive backup of live database to local timestamped directory:
mongodump --uri="mongodb+srv://<SOURCE_USER>:<PASSWORD>@<SOURCE_CLUSTER>/ashivam_technologies" --out=./backup_YYYYMMDD

# 2. Verify backup completeness and document counts:
ls -lh ./backup_YYYYMMDD/ashivam_technologies

# 3. Safe non-destructive restore into isolated NEW company-owned cluster (NO --drop):
mongorestore --uri="mongodb+srv://<TARGET_USER>:<PASSWORD>@<COMPANY_CLUSTER>/ashivam_technologies" ./backup_YYYYMMDD/ashivam_
technologies

# 4. Reconcile collection document counts in mongosh between source and target:
use ashivam_technologies
db.admins.countDocuments()
db.jobs.countDocuments()
db.inquiries.countDocuments()
db.applications.countDocuments()
```

11 Comprehensive REST API Reference

All endpoints operate under the `/api/v1` prefix with standardized JSON request/response envelopes. Every response echoes the originating `X-Request-Id` for distributed tracing.

11.1 Public Endpoints

HTTP METHOD	ENDPOINT PATH	AUTH	RATE LIMIT	PURPOSE & REQUEST PAYLOAD	EXPECTED RESPONSE
`GET`	`/api/v1/health`	None	Health Limiter	Diagnostic health probe inspecting MongoDB connection and uptime.	`200 OK` : `{ success: true, database: "connected", uptimeSeconds: 42 }`
`GET`	`/api/v1/health/liveness`	None	Exempt	Kubernetes / cloud container liveness check. Always returns fast 200 without DB query.	`200 OK` : `{ status: "ok" }`
`POST`	`/api/v1/inquiries`	None	10 / 15m	Submits client lead. Body: `{ name, email, phone, company, projectType, budget, message }`.	`201 Created` : `{ success: true, message: "Inquiry received" }`
`GET`	`/api/v1/career/jobs`	None	Global	Retrieves list of active, published job openings. Excludes draft/closed listings.	`200 OK` : `{ success: true, data: [ { title, slug, department... } ] }`
`GET`	`/api/v1/career/jobs/:slug`	None	Global	Retrieves detailed job specifications for a specific position slug.	`200 OK` : `{ success: true, data: { ... } }` or `404 Not Found`
`POST`	`/api/v1/career/apply`	None	10 / 15m	Multipart form submission: applicant fields + resume binary file (PDF/DOC up to 5MB).	`201 Created` : `{ success: true, message: "Application submitted" }`
`POST`	`/api/v1/auth/login`	None	5 / 15m	Admin login. Body: `{ username, password }`. Validates salt hash via bcrypt.	`200 OK` : `{ success: true, token: "JWT...", user: { role: "Superadmin" } }`

11.2 Protected Administrative Endpoints (Requires Bearer Token)

HTTP METHOD	ENDPOINT PATH	MINIMUM ROLE	PURPOSE & OPERATIONAL FUNCTIONALITY
`GET`	`/api/v1/inquiries`	Admin	Lists submitted client inquiries with search, status filtering, and pagination.
`PATCH`	`/api/v1/inquiries/:id/status`	Admin	Updates inquiry processing status (`in-review` , `contacted` , `archived`).
`POST`	`/api/v1/admin/jobs`	Admin	Creates a new career opening with title, department, requirements, and auto-generated slug.
`PATCH`	`/api/v1/admin/jobs/:id`	Admin	Updates opening specifications or toggles publication status (`published` , `closed`).
`GET`	`/api/v1/admin/applications`	Admin	Lists candidate submissions across all positions with status filtering.
`GET`	`/api/v1/admin/applications/:id/resume`	Admin	Streams candidate resume binary directly from S3 or local vault with correct Content-Type.
`GET / PUT`	`/api/v1/settings`	Superadmin	Inspects or updates global company contact info, social links, and maintenance state.

12 Authentication, Authorization & Security Hardening

12.1 Defensive Security Architecture

The platform implements deep defense-in-depth across client, transport, routing, and persistence layers:

SECURITY VECTOR	DEFENSIVE CONTROL IMPLEMENTED	CODE IMPLEMENTATION REFERENCE
Injection Defense	Strict payload schema validation, input sanitization, and Mongoose query parameterization preventing SQL & NoSQL injection.	`backend/src/utils/validation.js`
Session Cryptography	HMAC SHA-256 signed JSON Web Tokens (JWT) with 1-hour expiration; enforced ≥ 32-character secret key in production.	`backend/src/utils/jwt.js`
Password Storage	One-way bcrypt cryptographic hashing with salt rounds (12) defending against rainbow tables and brute force.	`backend/src/models/admin.model.js`
Spam & Bot Filtering	Silent honeypot field trapping automated bots on public forms without disturbing human users.	`isHoneypotTriggered()` in `validation.js`
DDoS & Brute Force	Multi-tier rate limiting: 5 logins / 15m, 10 forms / 15m, 100 global requests / 15m per IP with standard 429 response envelopes.	`backend/src/middlewares/rateLimit.middleware.js`
Malicious Uploads	Multer memory storage combined with file signature magic-byte audit verifying authentic PDF/DOC binary headers.	`validateFileSignature()` in `storage.service.js`
Data Exposure Prevention	Private upload routes (`/uploads/resumes`) explicitly blocked with 404 handler; error normalization redacts server stack traces in prod.	`backend/src/app.js` (lines 136-150)
HTTP Security Headers	Helmet integration: `X-Frame-Options: DENY` , `X-Content-Type-Options: nosniff` , `X-Powered-By` disabled.	`backend/src/app.js` (lines 34, 57-65)

13 Quality Assurance, Automated Testing & Verification Suite

Quality assurance is backed by 144 automated tests running natively on the Node.js test runner without third-party test framework overhead.

13.1 Test Execution Taxonomy & Actual Command Results

TEST SUITE CATEGORY	EXECUTION COMMAND & WORKING DIRECTORY	COUNT & DURATION	TAXONOMY & SCOPE COVERED
Frontend Static & Unit Suite	`npm test` *(Executed in root workspace)*	65 PASSED / 0 FAILED (828ms)	Component, Route & SEO Static Unit Tests: Route declarations, 44px touch targets, mobile drawer, responsive clamp typography, SEO meta, schema JSON-LD, content distribution.
Backend Unit & Integration Suite	`npm test` or `node --test tests/*.test.js` *(Executed in `backend/` directory)*	79 PASSED / 0 FAILED (5,858ms)	Unit & HTTP Integration Tests: Validation regex, honeypot traps, ReDoS escapes, JWT signing, supertest HTTP assertions for security headers, rate limiters, proxy trust, error envelopes.
Production Build Integrity	`npm run build` *(Executed in root workspace)*	CLEAN BUNDLE (280ms)	Vite production bundle generated into `dist/` (86 modules) without errors, warnings, or missing imports.
Whitespace Integrity	`git diff --check` *(Executed in root workspace)*	0 ERRORS (8ms)	Zero carriage return discrepancies or trailing whitespace errors across all tracked source files.

MOCKED VS. LIVE INFRASTRUCTURE TESTING & LINTER STATUS

The 79 backend automated tests execute against an in-memory/mocked HTTP application instance without requiring an active live MongoDB Atlas connection, AWS credentials, or SMTP server. Full end-to-end cloud database transactions, live S3 bucket streaming, and production email dispatches require active cloud accounts and are verified via staging deployments against live infrastructure.

Linter Configuration Status: No code linter (ESLint, Prettier, etc.) is configured in root `package.json` or `backend/package.json`. No linting command was run; claims of zero lint warnings have been removed.

14 Cloud Hosting & Production Deployment Blueprint

14.1 Render Cloud Backend Deployment Blueprint

- **Target Architecture Platform:** Render Cloud (Web Service) — *[Company account setup pending provisioning]*.
- **Infrastructure-as-Code:** `render.yaml` automatically provisions the service upon repository connection.
- **Service Settings (render.yaml):** Root Directory: `backend` | Plan: Starter | Region: Oregon | Build: `npm ci --omit=dev` | Start: `node src/server.js` | Health Check: `/api/v1/health` | Environment: Native Node runtime (`env: node`).
- **Container Alternative (Dockerfile):** Multi-stage `node:22-alpine` image, non-root user (`appuser`), port 5000 with `wget` health probe.
- **Zero-Downtime Transition:** Create the new Render service from `AshivamTechnologies/Official-Websites`. Verify health at `https://<company-backend>.onrender.com/api/v1/health` before pointing the frontend.

14.2 Vercel Frontend Deployment Blueprint

- **Target Architecture Platform:** Vercel (Production Team Project) — *[Company account setup pending provisioning]*.
- **Repository Connection:** Connect `AshivamTechnologies/Official-Websites` (branch: `main`).
- **Build Configuration:** Framework Preset: Vite | Root Directory: `./` | Build Command: `npm run build` | Output Directory: `dist`.
- **Build Variable:** Set `VITE_API_URL` to the new company Render backend URL.
- **Target Cloud Services:** MongoDB Atlas (database), AWS S3/Cloudflare R2 (storage), and Brevo/Resend (email) represent target architecture pending company provisioning and live integration verification.

15 Operational Maintenance, Troubleshooting & Handover Sign-off

15.1 Common Troubleshooting Scenarios

OBSERVED SYMPTOM	PROBABLE ROOT CAUSE	STANDARD RESOLUTION PROTOCOL
Backend boot failure: "JWT_SECRET must be at least 32 characters"	`JWT_SECRET` environment variable is missing or shorter than 32 characters in production mode.	Generate a 32+ character random string (e.g. `node -e "console.log(crypto.randomBytes(32).toString('hex'))"`) and set in Render dashboard.
CORS error on frontend during form submit	Frontend domain is not listed in backend `ALLOWED_ORIGINS`.	Add the Vercel domain or custom domain to `ALLOWED_ORIGINS` in Render dashboard and restart backend.
Uploaded resumes disappear after server redeployment	Backend is running with `STORAGE_PROVIDER=local` on ephemeral Render disk.	Configure AWS S3 or Cloudflare R2 credentials (`S3_BUCKET`, `S3_ACCESS_KEY_ID`, `S3_SECRET_ACCESS_KEY`) in `backend/.env` and execute `cd backend && npm run migrate:resumes`.
Contact form emails fail to deliver	Cloud platform blocks standard outbound SMTP ports (25/587).	Switch `EMAIL_PROVIDER` to `resend` or `brevo` and configure the corresponding HTTPS API key in Render dashboard.

15.2 Handover Verification & Sign-Off Checklist

VERIFICATION ITEM	TARGET STATE	VERIFIED STATUS
Company GitHub Ownership	`AshivamTechnologies/Official-Websites.git` (Commit `6b2fae0`)	VERIFIED MATCH
Handover Backup Integrity	`C:\Users\DELL\Desktop\Official-Website-Handover-Backup`	VERIFIED PRESENT
Frontend Automated Suite	65 passing static & unit tests (routes, 44px touch targets, mobile drawer, responsive clamp typography, SEO meta, schema JSON-LD).	65 PASSED / 0 FAILED
Backend Automated Suite	79 passing unit & HTTP integration tests with mocked transports (validation, auth guards, rate limiters, security headers). In-memory mock tests only; live MongoDB/S3/SMTP not executed.	79 PASSED / 0 FAILED (MOCKED)
Production Client Build	Vite production compilation bundling 86 modules into `dist/` in 280ms with 0 errors or warnings.	VERIFIED CLEAN
Whitespace Integrity	`git diff --check` across all tracked source files; 0 carriage-return discrepancies. (No code linter configured in `package.json`).	0 ERRORS (NO LINTER)
Candidate Resume Files	24 sample candidate resumes confirmed present in local `uploads/` (untracked & protected).	VERIFIED PRESENT
Live Personal Backend & DB	Personal Render deployment & MongoDB Atlas cluster. Real-time health probe timed out during audit; status not independently verified.	NOT INDEPENDENTLY VERIFIED
Company Cloud Hosting (Target)	Target architecture: Render, Vercel, MongoDB Atlas, S3/R2, and Brevo company accounts.	PENDING PROVISIONING
Handover Documentation	Complete 15-chapter comprehensive PDF manual & Markdown source.	COMPLETE

Engineering Handover Sign-Off

This technical document confirms that the Ashivam Technologies Official Website application source code, automated regression test suites, build pipelines, and Git history have been audited, stabilized, and synchronized to company repository ownership at commit `6b2fae02b0aca4138fb4ecb77d7b7cfe744ae408`. Company cloud hosting infrastructure provisioning and live cloud integration testing remain pending.

Engineering Specialist: Lead DevOps & Systems Architect **Handover Date:** October 16, 2026

Handover Scope: Source Code Synchronized (Commit 6b2fae0) • **Cloud Status:** Company Cloud Infrastructure (Target Architecture) Pending Provisioning